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THE PLACE OF DIESEL ENGINES IN MODERN SOCIETY

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Technical Communication Techniques

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ILLUSTRATIONS

Introduction

Before the advent of the engine, man weaved material and harvested crops by hand and travelled either by foot, or horse and carriage. Productivity increased when horses, waterwheels, and wind were used to drive small machines. Inventions such as the spinning jenny and flying shuttle increased textile manufacture but the steam engine completely revolutionised all aspects of industry.

Steam engines increased productivity and were a major contributor to the Industrial Revolution in the eighteenth century. Steam engines also revolutionised transport; people and goods could travel at unprecedented rates. These engines used coal to heat water outside the engine, and the thermal energy in the steam caused the parts of the engine to move. Diesel engines were used in pumps, locomotives, steamships and steam tractors, but were inefficient as they often needed two people to operate; for example, a steam engine needed a driver and someone to feed the fire.

A significant development in engine design was when the German engineer Nikolaus August Otto developed the internal-combustion engine which used fuel and air inside the engine, as opposed to outside. A spark plug ignited the fuel and air within the cylinder, and the combustion caused the pistons in the engine to move. The internal-combustion engine used petrol as opposed to coal, and by the early 1900s was the most common engine which powered motorcycles, automobiles and small trucks.

Another German engineer, Rudolf Diesel, realised ignition was not needed in an internal-combustion engine if the air within the cylinder was compressed, which would cause it to rise in temperature. Fuel injected into the cylinder would then ignite. He developed the first diesel engine in the early 1890s which used heavy oil, a cheaper and safer alternative to petrol.

The petrol engine is the most common engine in automotive vehicles today, particularly cars. Previously diesel engines have been regarded as noisy and smelly, and not capable of the rapid acceleration required in cars, particularly 'sporty' models. This report looks at the development and general principles of the diesel engine, and how they compare with petrol engines. This report also aims to show what effects recent developments such as fuel injection and turbo charging have had on modern diesel engine performance when compared with petrol engine performance.

General Engine Design and Development

The term engine refers to any mechanical device that operates machinery, generates electricity, pumps water or compresses gas. An engine is powered by the combustion (ignition) of fuel. **Fig. 1** explains the characteristics of the two types of combustion engine, external and internal. Engines can run on either two or four-stroke cycles; this number relates to the number of stages required to convert fuel into a force that can power the machinery. Engines that operate on the four-stroke cycle are more popular today as they provide more power than the two-stroke cycle. Therefore, this report uses the four-stroke engine cycle as an example.

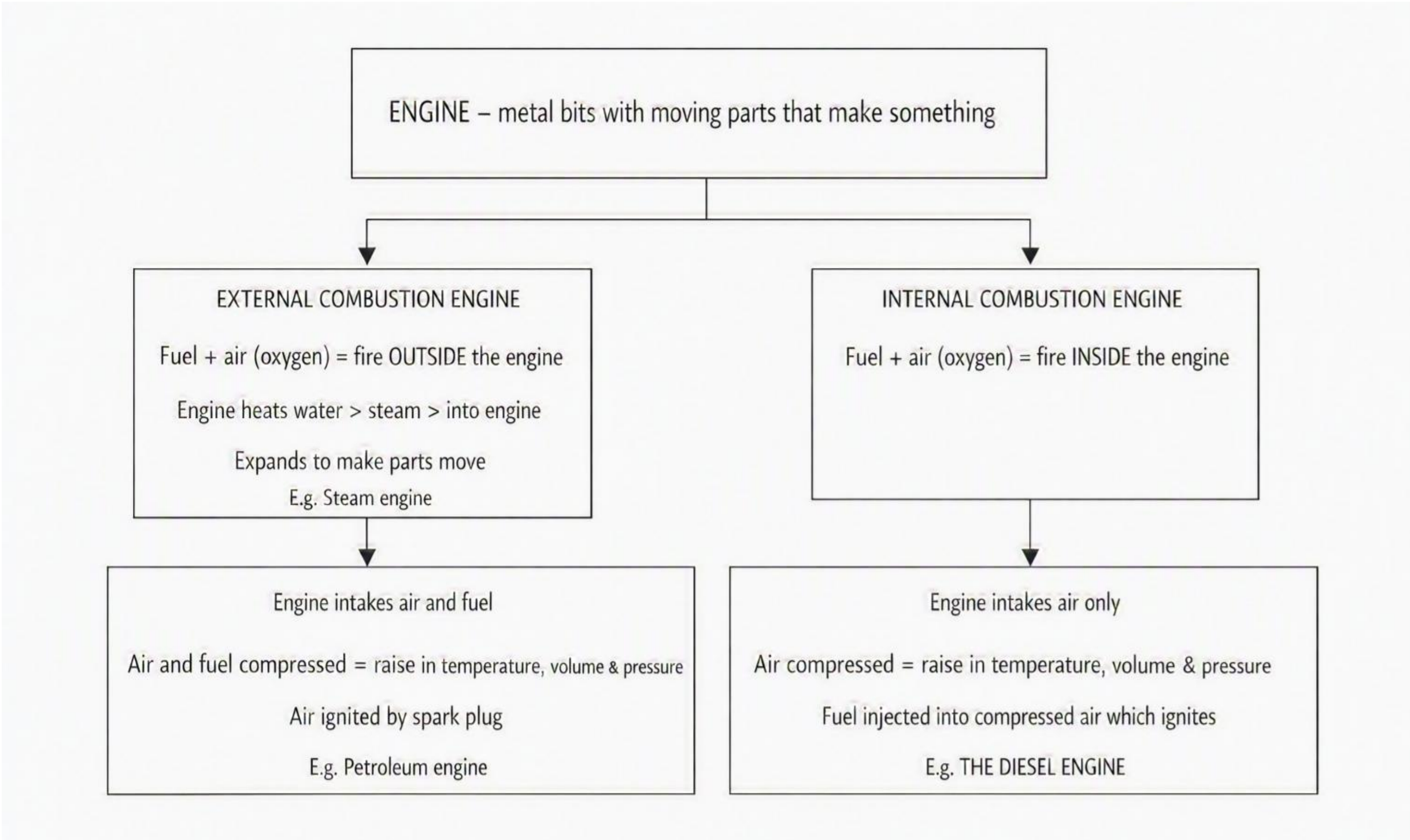

Figure 1 - Characteristics of External and Internal Combustion Engines

Fig. 2 shows the major components in an engine. The valves in a petrol engine open to let in air and fuel, whereas a diesel engine only takes in air. The gas and fuel in a petrol engine is compressed, and a spark plug ignites the mixture which causes it to combust. The air in a diesel engine is compressed until it reaches a temperature that will cause fuel injected into the cylinder to ignite. In both types of engines, the gases that are a result of the burned fuel raise the temperature and pressure within the engine. This increase causes the parts of the engine to move; the piston, a cylindrical piece of metal, moves down the cylinder and pressure forces it up again.

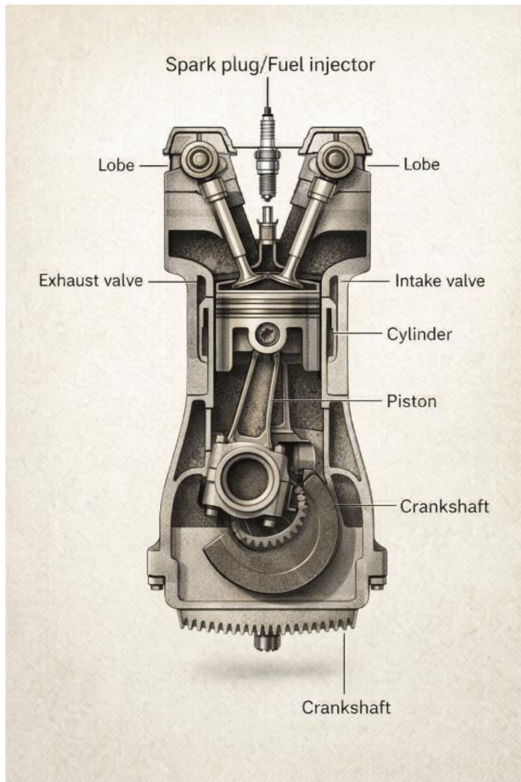

Figure 2 - Major Components in an Engine

The crankshaft then converts the up and down movement of the piston into a circular motion, using the same action as pedals on a bike. **Fig. 3**, drawn from above, illustrates this principle. The clutch and gearbox are then given power and control the behaviour of the engine. A valve opens to let out the exhaust fumes, the result of the fuel combustion. Both the intake and exhaust valves are controlled by oval lobes; their shape forces the valves up and down at the correct moment.

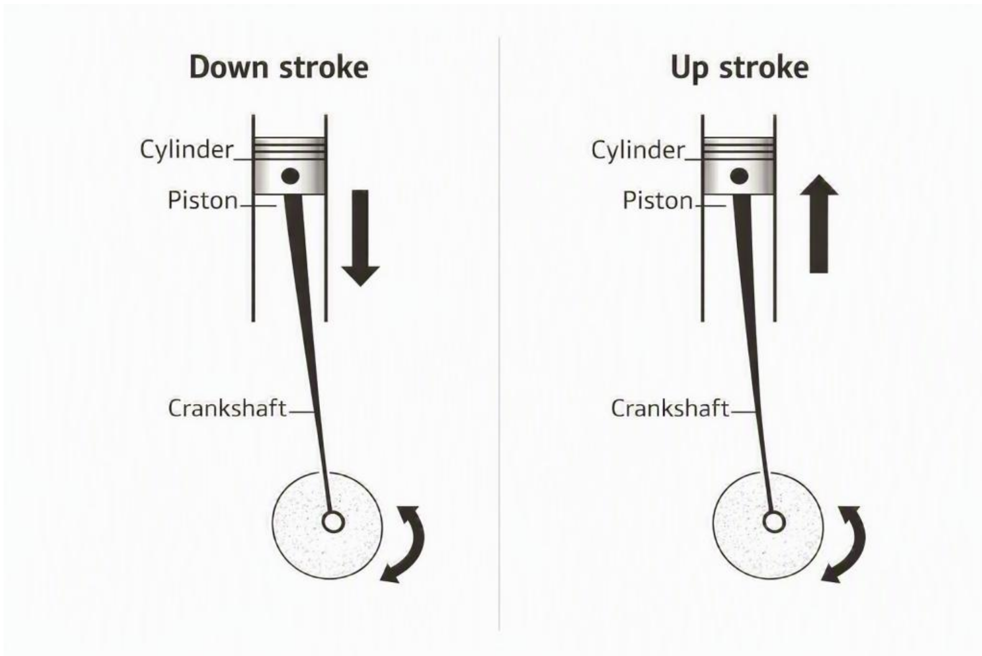

Figure 3 - Conversion of the Piston Movement from Up to Down to Circular

Development of the Diesel Engine

The original diesel engine used coal ground into dust as fuel. Coal dust was used as opposed to chunks of coal because it has a greater surface area per unit weight and was therefore more likely to combust. Compressed air fed the coal dust into the cylinder, but it was difficult to control the rate of injection. This caused Diesel's experimental engine to be destroyed in an explosion. From then on, Diesel used liquid petrol (oil) in his engines and continued to introduce the fuel into the engine with compressed air.

Diesel had no business talent and did not profit from his invention. He died in 1913 in the North Sea after allegedly falling from a ship. He was about to meet with representatives of the motor company Rover and was allowing anyone to buy a licence for his engine patents. This included companies in France, Britain and other nations who were enemies of Germany. German agents may have murdered Diesel to prevent him from licensing further patents to other foreign companies. World War I was about to begin.

General Principles Behind the Diesel Engine

The diesel engine uses the principles behind Charles's law, Boyle's law, and Gay-Lussac's law, which state that when a gas is compressed its temperature, volume and pressure increase respectively. Air is drawn into a cylinder, and a rising piston compresses it at a ratio between 18-22:1, which raises its temperature to 700-900°C, or 1300-1650°F, sufficient to ignite fuel. Fuel is injected at the top of the piston stroke as finely atomised spray, furthest from the crankshaft (also known as the TDC, or top dead centre). The fuel is injected as a spray, so the greatest surface area of the fuel is subjected to the air, which produces the greatest amount of power. The piston is forced downwards when the fuel combusts. The connecting rod then forces the crankshaft to turn, which delivers rotary power at the end of the crankshaft. The piston pushes the exhaust gas up the cylinder, and valves open to let the exhaust gas out, and draw in fresh air.

When a diesel engine is cold, the temperature of the compressed air in the cylinders may not be high enough to ignite the fuel. To overcome this, glow plugs heat the air in the combustion chamber just before and during start-up.

A Comparison of Automotive Diesel and Petrol Engines

The main difference between a diesel and petrol engine is how the fuel and air mixture is introduced into the cylinder and ignited. A petrol engine mixes the fuel with air before it enters the cylinder, and the mixture is then ignited at the correct moment by a spark plug. A diesel engine draws only air into the cylinder which is then compressed and fuel injected as a fine spray as the piston reaches the end of the compression stroke, which causes the fuel to ignite. Diesel engines are more fuel efficient because exactly the right amount of fuel is injected for combustion to occur. Diesel engines also have the advantage of burning the fuel more completely than petrol engines; more of the fuel is used.

Diesel engines have the advantage of lower fuel consumption and can be run with cheaper fuel. Consequently, diesel engines contribute less to the depletion of non-renewable fossil fuels.

Diesel engines are more reliable than petrol engines because they do not contain an electrical ignition system. Diesel engines are heavier than petrol engines as they are built to withstand the strain of the internal pressure needed for the fuel to combust. Consequently, diesel engines are more expensive to manufacture. However, their robust design ensures a longer working life; it also means diesel engines can withstand turbo charging which results in an increase of power. However, any modifications could result in a shorter life and a greater need for regular engine services. Petrol engines are more popular than diesel engines mainly because they are cheaper because they are not as robustly made.

Diesel engines cannot accelerate as quickly as petrol engines, even when turbo charged, because they are so large and heavy. **Fig. 4** illustrates a greater force is needed to compress the gas in a diesel engine, as opposed to a petrol engine, before it will ignite. Therefore, a diesel engine is made up of stronger components and is more robustly made. Once a diesel engine has accelerated, fuel injection and turbo charging can ensure it will perform to the same standard as a petrol engine.

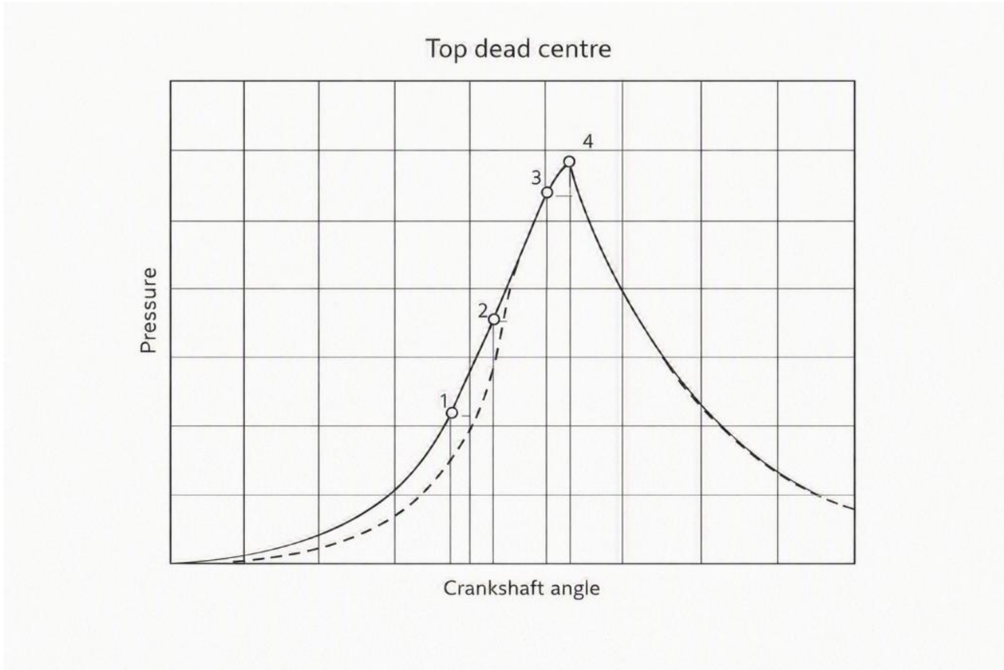

Figure 4 - Variation of Pressure in a Diesel Engine Cylinder

Fig. 4 shows the variation of pressure within a diesel cylinder as the crankshaft turns and moves the piston up and compresses the gas. The solid curve indicates the performance of a petrol engine, where the fuel is injected before the combustion takes place. The dotted line represents the lower pressures found if air alone is compressed, as in the diesel engine. Point 1 is the start of injection; point 2 is the start of combustion; point 3 is the finish of injection and point 4 is the finish of combustion. This graph shows that more energy is needed to pressurise the gas in a diesel engine, as opposed to the fuel and gas mixture in a petrol engine. Therefore, the diesel engine requires more power to start but once started provides power at low revs and greater fuel efficiency. Its combustion cycle also creates less violent changes in pressure when compared with a petrol engine.

One drawback of diesel engines is their emission of air pollutants. These engines typically emit high levels of soot from the exhaust, which consists of unburned carbon compounds. Whilst in older diesel engines this was a problem, modern diesels now catch this soot in a particle filter, which when saturated is automatically regenerated by burning the caught particles. However, it is worth noting that modern diesel engines produce smaller soot particles than older models, some of which are too small to be caught by the filters. Therefore, the older models were less harmful to humans and the environment. Diesel engines also emit reactive nitrogen compounds, and high levels of odour when compared with petrol engines. However, catalytic converters and exhaust gas recirculation systems have become standard on most modern diesel engines to meet increasingly stringent emission regulations. Modern advances in diesel engine design ensure that exhaust emissions are filtered so they are no louder than a petrol engine.

'Knock' and Smoke in Diesel Engines

In the past diesel engines were considered to be noisy and smoky, and this was true to some extent. It is worth looking at the causes of knock and smoke to see how they have been reduced in modern engines, and to understand what makes them worse. Knock is caused by a small delay (typically 0.001 to 0.002 second) between the start of fuel injection and combustion. This delay, known as 'ignition lag', is greatest when the engine is cold and idling. The knock is caused by the sudden increase in cylinder pressure when the injected fuel has been mixed with the hot air and ignites. It is therefore an unavoidable part of the combustion process but has been reduced by improvements to the combustion chamber and fuel injection design.

Smoke is caused by incorrect combustion. Thick black smoke indicates a lack of air combustion, either because the air intake is restricted (clogged air cleaner), or because too much fuel is being injected (defective injectors or pump). Unlike knock, smoke is preventable by cleaning the air cleaner or replacing any defective engine components. During start-up and warm-up some white or blue smoke may be seen, but under normal running conditions the exhaust should be clean.

The Diesel Four-Stroke Cycle

Diesel engines use either a two-stroke or four-stroke cycle, the latter is more commonly used as it uses fuel more efficiently, burns fuel cleaner and provides greater power (torque) at low speed. However, the four-stroke engine requires a greater number of moving parts and manufacturing expertise and is larger and heavier when compared with the two-stroke engine of comparable power. In the typical four-stroke-cycle engine, the intake and exhaust valves and the fuel-injection nozzle are in the cylinder head. These engines often contain two intake and two exhaust valves.

Fig. 5 illustrates the following steps in the diesel four-stroke cycle:

1. *Intake***:** the piston comes down, the valve opens and air is drawn in.
2. *Compression*: the piston moves up which compresses the air in the cylinder.
3. *Power (ignition)*: the compressed air increases in temperature, pressure and volume and ignites fuel injected into the cylinder. The piston moves down and the crankshaft rotates. This is the most important step of the cycle as power is generated which drives the engine.
4. *Exhaust*: the valve opens and fumes from the ignited fuel are pushed out of the cylinder (scavenged). Air is drawn in ready for the next piston cycle.

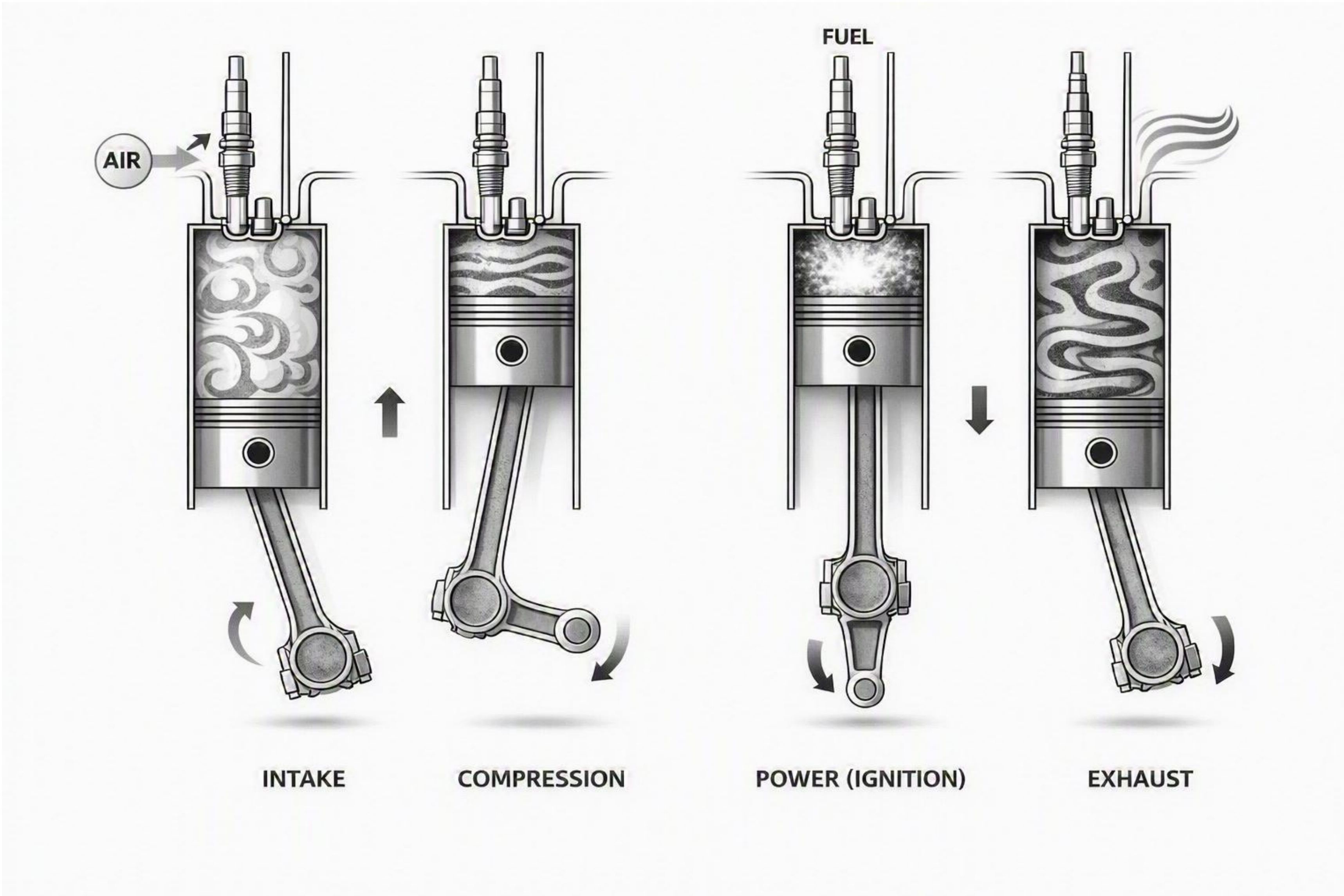

Figure 5 - The Diesel Four-Stroke Cycle

Fuel Injection in Diesel Engines

Precise control of the fuel injection is critical to the performance of a diesel engine because it controls the entire combustion process. One fuel injector is fitted to each cylinder in the engine. Fuel injection distributes the fuel evenly to the individual cylinders, which develops power, increases fuel economy and reduces pollution levels. Fuel is sprayed so the greatest surface area of the fuel is subjected to the pressurised air. Injection must begin at the correct piston position, or crank angle, to avoid ‘knocking’, when the unburned mixture in the combustion chamber explodes before the flame can reach it, and it combusts before the peak of the cycle. At first, the fuel burns in nearly a constant volume while the piston is near the top dead centre. The fuel continues to be injected as the piston moves away from this position, which provides the combustion process with constant pressure. A governor regulates the fuel delivery to control the speed of the engine; when the fuel pressure exceeds a certain value, the flow of fuel is stopped.

Fuel can be introduced by either direct or indirect injection. **Fig. 6** shows direct injection, in which fuel is delivered directly into the main combustion chamber. There are two types of fuel injection pump which are determined by engine size.

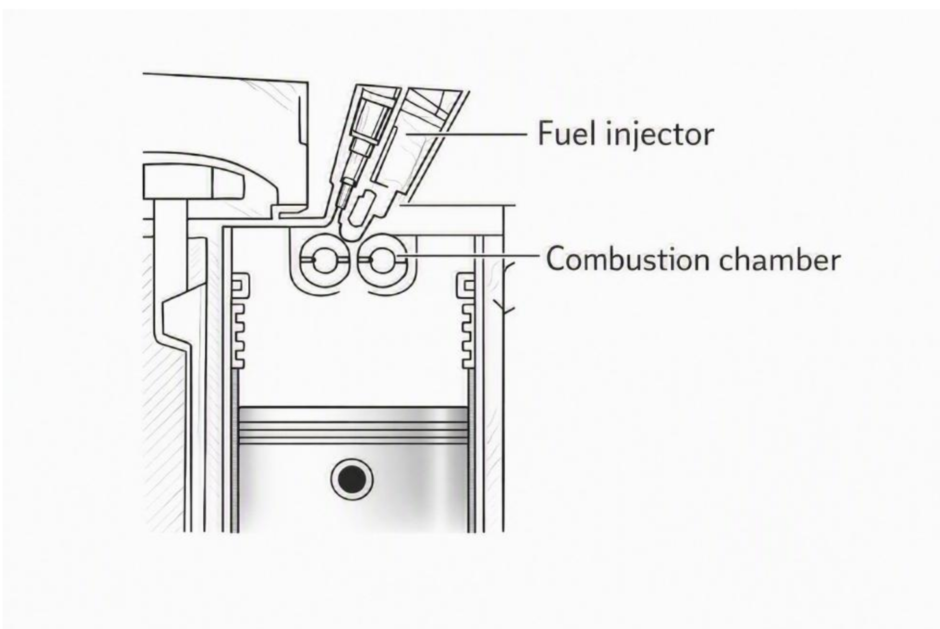

Figure 6 - Direct Injection

Fig. 7 shows indirect injection, in which fuel is delivered to a separate pre-combustion chamber. Combustion then begins and spreads into the main combustion chamber. The pre-combustion chamber is designed to ensure the fuel mixes sufficiently with the compressed air, which slows the rate of combustion. This results in a reduction of noise, softer shock of combustion and lower stresses on the engine components. One drawback to indirect injection is the pre-combustion chamber causes increased heat loss which reduces the efficiency of the engine.

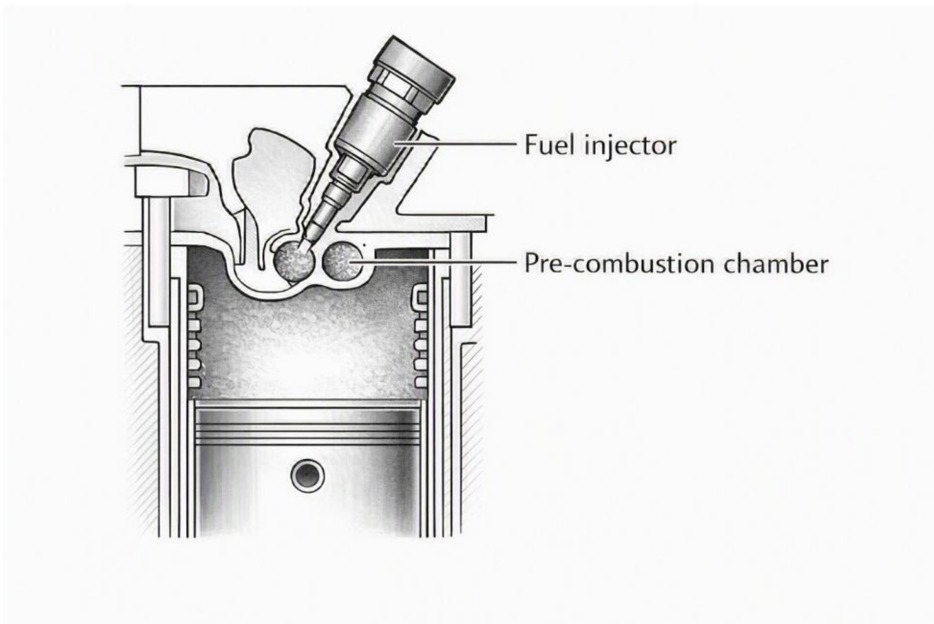

Figure 7 - Indirect Injection

Small engines use rotor pumps, where one plunger pumps fuel to each of the cylinders. Larger engines use a pump with plungers that operate simultaneously with the same volume of fuel. The rotor pump cannot hold much fuel, and consequently in-line pumps are used on larger engines, where there is one pump cylinder for each cylinder.

Electronic Diesel Engine Control Systems

Electronic diesel engine control systems have been developed in recent years to improve diesel engine efficiency and reduce exhaust emissions. These systems are reliable because they are solid-state devices that contain no moving parts. They ensure correct amount of fuel is injected at the correct pressure, at exactly the right time. Almost all modern engines use some form of electronic engine control system. They perform an essential function because even small variations in fuel supply can cause increased exhaust emissions, and increased noise and fuel consumption. In a typical diesel engine, the injection process takes only a thousandth of a second, and only a small amount of fuel is injected. A conventional mechanical fuel injection pump uses an accelerator cable connected to the driver's pedal with various other devices such as cold start injection advance and fast idle units. However, even with these add-on devices, it is difficult for a mechanical diesel control system to keep up with the modern demands on engine refinement and exhaust emission control.

Many electronic diesel engine control systems use a conventional in-line or distributor fuel injection pump, with the injection pump timing and the quantity of fuel injected controlled electronically rather than mechanically. **Fig. 8** shows the electronic sensors in an electronic diesel control system. The sensors measure the accelerator pedal position, engine crankshaft speed, engine camshaft position, the mass of air passing into the engine, turbocharger boost pressure, engine coolant temperature and ambient air temperature.

Information from the sensors is passed to an electronic control unit (ECU), which evaluates the signals. The memory of the ECU contains mapped values for injected fuel quality and start-of-injection point. The ECU performs calculations based on the information provided by the sensors and selects the most appropriate values for the fuel quantity and start-of-injection point from its stored values. The ECU can analyse the data and performing calculations many times per second, which allows accurate control of engine operation.

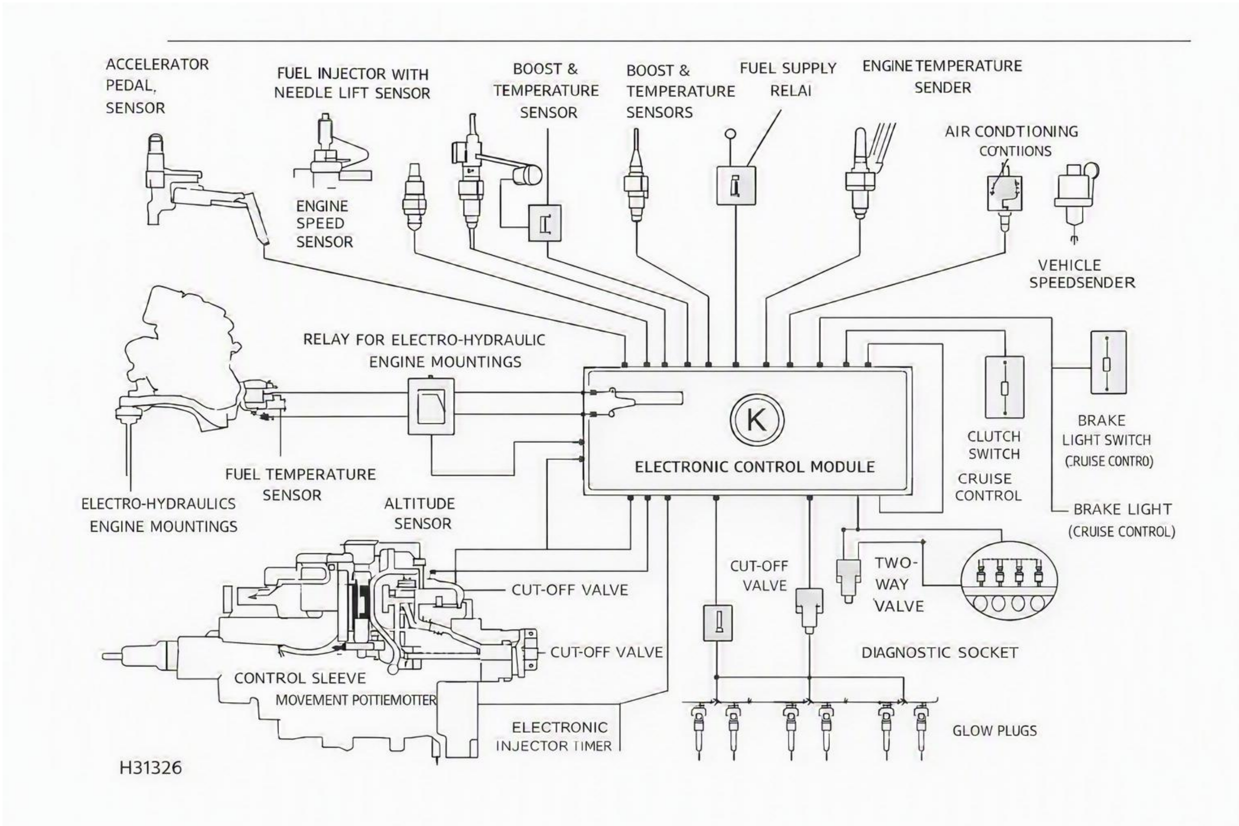

Figure 8 - Typical Electronic Diesel Control System Components

Turbo Charging in Diesel Engines

A turbocharger uses energy from the exhaust gas to compress and force air into the engine's cylinders. The air is cooled before going in to the engine because cooled air is denser and contains more oxygen in a given volume than warm air. Fig. 9 illustrates that power output and torque (the force that causes rotation) in a diesel engine increase when the air is cooled. A key advantage of turbochargers is that they offer a considerable increase in engine power with only a slight increase in weight. Turbochargers are a disadvantage in petrol engines as the compression ratio needs to be lowered so the maximum compression pressure is not reached, and to prevent engine knocking. However, this disadvantage does not apply to engines that are designed to be turbocharged.

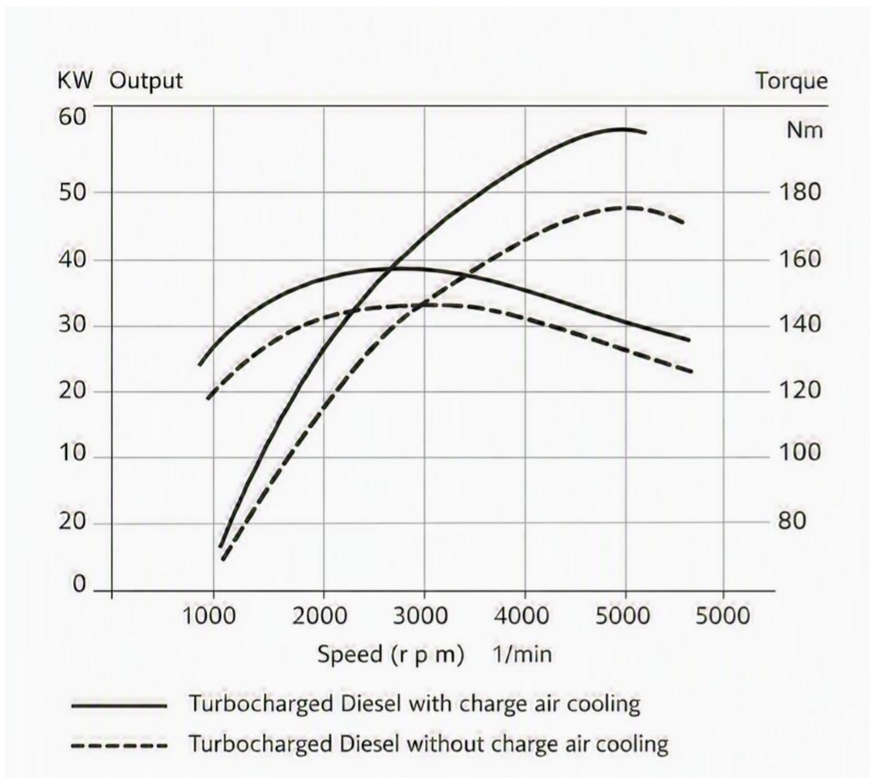

Figure 9 - Power and Torque Output from a Turbocharged Engine with and without Air Cooling

Most modern diesel engines are turbo charged to increase their power and performance. Diesel engines are suitable for turbo charging for several reasons. Diesel engines have lower power-to-weight ratios when compared with petrol engines, and therefore their power increases greatly when turbo charged. Diesel engines are robustly built because they run under very high levels of pressure and temperature and can therefore cope with the addition of a turbocharger. The air intake can be subjected to greater pressure by the turbo charger because it does not contain fuel and air as with a petrol engine.

Applications of the Diesel Engine

The diesel engine uses the least amount of fuel of any large internal-combustion engine and is consequently the most economical. Diesel engines are also heavyweight, robustly designed, and provide an efficient rate of power (torque) at low revs. Consequently, diesel engines are used in machinery involved in constant heavy-duty jobs including large marine engines, modern heavy road vehicles such as trucks, ships, large-scale portable power generators, most farm and mining vehicles, and many long-distance locomotives. They are also used in hospitals, telephone exchanges, and airports to provide emergency power during electrical power cuts.

The zeppelins Graf Zeppelin II and Hindenberg were the two largest aircraft ever built and provided passenger transport as well as military support during the two World Wars. They used reversible diesel engines whose propellers were driven directly from the engine. To go in reverse, the engine was stopped from going forwards, and the propellers set in the opposite direction by shifting gears on the camshaft. They could be brought to full power in reverse in less than 60 seconds. This was before reversible pitch propellers for aircraft had been perfected.

In 2001, nearly 36% of newly registered cars in Western Europe had diesel engines. Diesel engines can be divided into three sizes; small, medium, and large, directly related to the size of the cylinders within the engine. Small diesel engines (approximately 1200 – 3500 rpm) power trucks, buses, tractors, cars, yachts, compressors, pumps, and small generators. They can also power small stationary electrical power generators such as those on pleasure craft, and as mechanical drives. Some motorcycles use small diesel engines.

Medium engines (approximately 300 to 1200 rpm) power heavy-duty trucks and large electrical generators, run at a set speed and provide a rapid response to load changes. Large diesel engines (approximately 60 to 100 rpm) power ships and locomotives, to generate electrical power, and for mining and construction. They often run on cheap, low-grade fuel which needs extra heat treatment in the ship for tanking and before injection because of their low volatility.

Submarines are propelled forward by electric motors run from energy stored in batteries charged by a generator run by a diesel engine. The diesel engine is run when the submarine is on the surface, which charges the batteries that provide it with power when submerged. Almost all large marine vessels are powered by diesel engines, as they allow greater range at lower cost, and are more powerful at lower speeds than petrol engines.

The weight and lower output of diesel engines mean they are not used for automotive racing. They are, however, used in truck racing and types of racing where these drawbacks are less severe, such as land speed record racing. Some dragsters are powered by diesel engines, despite the drawbacks being central to performance in this sport. This will undoubtedly change in the future as modern diesel engines are lighter and accelerate more rapidly than their predecessors.

Modern diesel cars have been engineered and developed and consequently the difference between petrol and diesel automotive vehicles has significantly narrowed. Large, successful companies such as Audi and Ford develop and sell many vehicles with diesel engines, such as the Audi 3.0 TDI Quattro, which are modified to provide high performance. **Fig. 10** shows how diesel engines are used in modern society; for example, in cars, tractors, trains and submarines.

The common rail diesel injection system is a recent development in electronic diesel engine control. In this system a common fuel reservoir is used to supply fuel to the injectors. Common rail systems allow fine engine control, and reduce the noise and harshness often associated with direct injection engines.

Figure 10 - Examples of how the Diesel Engine is used in Modern Society

Conclusion

The original diesel engines were heavy and cumbersome and produced a great number of emissions when compared with petrol engines. However, modern advances in turbo charging and fuel injection, including electronic engine control systems, have greatly reduced these emissions, and improved the performance and fuel economy of the diesel engine.

For many years, diesel engines played a vital role in modern society, particularly in commercial transport, agriculture, construction, and industry, where their durability, efficiency, and high torque characteristics made them the preferred choice. Although increasing environmental concerns and the growth of electric and hybrid vehicles have led to a reduction in diesel use in some sectors, diesel-powered engines continue to remain essential in many applications where alternative technologies are not yet practical or economical.

The growing adoption of electric and hybrid vehicles has also influenced the role of diesel engines in modern society. While electric vehicles are increasingly common in passenger transport, diesel engines continue to provide advantages in sectors such as heavy goods transport, agriculture, construction, shipping, and power generation, where high torque, durability, and long operating ranges remain essential requirements.

The future of diesel technology is likely to be shaped by further improvements in emissions reduction, the development of renewable and synthetic fuels, and the ongoing transition towards lower-carbon transport systems. As a result, the diesel engine continues to occupy an important, albeit changing, place in modern society.

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Multimedia

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Summary

This report explains the general principles and development of diesel engines. A comparison of diesel and petrol automotive engines shows how developments in modern diesel engine design have narrowed the differences between the two types of engines. The report also explains the variety of ways in which diesel engines are used in modern society, and the reasons behind these uses.

The background knowledge necessary to appreciate the contents discussed in this report would be within the capabilities of anyone with school leaving qualifications such as ‘O’ level or GCSE physics and chemistry.

The symbols used in the functional diagram are specified in BS 3939: 1991 Specification of Graphical Symbols for Electrical and Electronic Technology.